Artificial Intelligence-generated Metadata in Library and Information Centres: The Case of ChatGPT

Sugabsen Martins[1]

Ukawuba Vivian Chinazo[2]

[1]Abdurrahman Ghaji Library, Adamawa State University, Mubi

[2]National Root Crops Research Institute, Umahia, Abia State

Abstract

In recent years, Artificial Intelligence (AI) has become an essential tool in various fields, including Library and Information Centres (LICs). It is very important for library services to create metadata, which is structured information that describes, explains, locates, or otherwise makes it easier to find, use, or manage an information resource. This study was conducted to identify how accurate and consistent ChatGPT provides metadata for books and also identified the challenges associated with using ChatGPT. To achieve the objectives, The ChatGPT 4o open access part was used. Three books were used to craft queries that prompted ChatGPT to provide metadata as well as MARC records for the books. It was found that ChatGPT provided accurate and consistent metadata and MARC records for the information resources identified; it gives more metadata elements of books that have e-versions online. The challenges associated with using ChatGPT are inefficiency in a fluctuating network environment and its inherent bias to books that are not present online. It was recommended, among other things that librarians should not depend absolutely on using ChatGPT in cataloguing and classification of information resources; ChatGPT should be used in a strong high bandwidth network environment and; a separate query should be made when in need of call number for books.

***Keywords*:** ChatGPT, Metadata, Artificial Intelligence, Library

Introduction

Libraries have historically functioned as institutions that store and preserve a vast array of information and knowledge (Olutoki & Osoba, 2017; Stepanov, 2019). Libraries are undergoing adaptations in response to the progression of technology in order to enhance the quality of their services and facilitate more accessibility for their patrons (Williams, 2021).

Managing AI-generated metadata in library and information centers refers to the systematic process of creating, evaluating, refining, integrating, maintaining, and ensuring the interoperability of metadata generated by artificial intelligence (AI) tools and techniques. This process aims to optimize the quality, accuracy, and consistency of metadata, enabling effective discovery, accessibility, and management of digital resources in library and information centers.

The increasing use of Artificial Intelligence (AI) in generating metadata has transformed the landscape of libraries and information centers. AI-generated metadata offers improved efficiency and accuracy but also raises concerns about data quality, consistency, and management. Effective management of AI-generated metadata is crucial to ensure discoverability, accessibility and preservation of digital resources.

The utilisation of Artificial Intelligence (AI) and machine learning is a burgeoning technological advancement with the capacity to revolutionize the operational dynamics of libraries (Yusuf, et al.,2021; Sonawane & Chavan, 2020). These technologies have the potential to automate the labour-intensive process of generating metadata and enhance the efficiency of delivering library services. Metadata plays a crucial role in enhancing the discoverability and accessibility of resources within a library's collection by providing significant descriptive information (Musoke, 2022). Nevertheless, the process of metadata development has conventionally relied on manual labour, requiring significant effort and resources (Park & Lu, 2020). The task of manually creating high-quality metadata becomes increasingly difficult to manage due to the steady influx of new

resources obtained by libraries. Yang et al. (2019) pointed out that, artificial intelligence (AI) and machine learning have the potential to provide automated solutions for metadata production. This can result in cost reduction and the allocation of staff resources towards more advanced tasks. Artificial intelligence techniques, like natural language processing Aria, GPT, Bard, Anthropic, etc., can extract essential information from documents and generate metadata entries automatically (Kolawole, 2021). Machine learning algorithms can train using pre-existing metadata, making predictions based on criteria like title, originator and subject. Artificial Intelligence (AI) is transforming library services, providing unparalleled prospects for enhancing efficiency, accessibility, and user experience. This research paper offers a thorough examination of the accuracy and consistency of artificial intelligence (AI) technologies in libraries. It explores the advantages, difficulties and potential future developments of using AI in libraries. The adoption of AI in libraries brings forth numerous advantages. It optimizes operational efficiency by automating routine tasks, minimizes human errors, and facilitates round-the-clock access to information resources.

Statement of the Problem

The proliferation of digital resources in libraries and information centres has underscored the need for accurate metadata to enhance resource discovery and accessibility. Traditional metadata creation methods can be labour-intensive and prone to inconsistencies. With the advent of artificial intelligence, there is growing interest in leveraging AI-generated metadata to streamline this process. However the efficiency and effectiveness of AI-generated metadata in improving the discoverability, accuracy and overall utilization of library resources remain under-explored.

This research aims to investigate how accurate and consistent AI-generated metadata are in libraries and information centres. By examining the accuracy and consistency, this study seeks

to provide insight into its effectiveness and limitations ultimately guiding future practices and technologies of metadata in libraries and information centres.

Objectives of the Study

i. To ascertain the accuracy of AI-generated metadata

ii. To ascertain the consistency of AI ingenerating metadata.

iii. To identify challenges associated with AI-metadata.

Review of Related Literature

Accuracy of AI Generated Metadata

Artificial Intelligence (AI) can be effectively and efficiently used in managing bibliographic records of information resources in libraries. This can be attained if it can maintain accuracy and consistency each time it is prompted about a particular information resource. In a study on Artificial Intelligence-assisted classification of library resources: the case of Claude AI by (Martins 2024), it was found that Claude, a generative AI can generate call mark of a given information resource with detailed taxonomy by analyzing the title of the book or the particular information resource in question however, subsequent prompting of the AI with the same query returns slightly different result each time. This indicates its inability to be accurate all the time.

The study concluded that AI can reduce intense mental work when performing cataloguing but it is not automatic as some people may think because the AI displays inaccuracies and inconsistencies in providing call number of books. In another study (Wada et al., 2023) performed a study with the theme Artificial Intelligence (AI) and Machine Learning for Metadata Generation and Library Service Delivery: A Systematic Review. Their findings show that AI-powered chatbots and virtual assistants are bringing transformation into library work even though the human factor

has not been displaced. The study concluded that AI has greater prospects for the 21st century learning community. In another study, (Mendez et al., 2020) posits that AI-generated metadata can enhance the discoverability of library resources by improving keyword generation and providing more detailed content descriptions. According to (Brown & Wilson, 2023), academic institutions, such as the Massachusetts Institute of Technology (MIT) Library explored the use of AI models for metadata generation. These systems have been particularly useful in managing the growing number of research papers and theses in institutional repositories. In another study carried out by (Cocciolo, 2025) with the theme oral history metadata: a study from an LGBTQ+ archival context found that 51.2% of the time respondents prefer AI-generated description compared to descriptive metadata created by human beings 22.6% of the time the reason being that human created metadata is brief compared to the AI generated metadata.

Consistency of AI in Generating Metadata

The three main focuses of AI are perception, reasoning, and action. Perception involves the ability of computers to interpret and understand sensory inputs from the environment, such as images or speech. Reasoning is fundamental for intelligent decision-making and problem-solving, involving internal processes or algorithms that enable computers to determine the best course of action (Ali & Owoeye, 2022). Artificial Intelligence is primarily focused on developing intelligent computers that can do tasks in a way that imitates human intellect. Whether strong or weak, AI systems aim to mimic aspects of human cognition to solve complex problems and adapt to new situations effectively (Tyring cited in Hernandez-Orallo 2017). In a study by (Omame & Alex-Nmecha, 2020) on Artificial Intelligence in Libraries: Prospects and Challenges for Nigerian Academic Libraries. Their work was centered on how and in which areas in the library artificial intelligence could be applied. They noted that AI can be applied in library and information centres in the area of acquisition cataloguing, indexing using expert systems, Natural Language Processing

(NLP), information retrieval and recommendation systems. It is in-line with this that this study is seeking to discover whether ChatGPT perform with consistency in generating metadata for information resource in libraries and information centre. Writing on the benefits of AI in libraries (Wu et al., 2021) hold that unlike humans, who may introduce inconsistencies or errors, AI models can generate metadata that adheres to a consistent format and is free from typographical or semantic errors. In his study on the theme can ChatGPT evaluate research quality? (Thelwall, 2024) showed that in providing index, descriptors, subject headings and class numbers ChatGPT is not consistent and therefore needs more training in order for it to be consistent.

Challenges Associated with Using AI Metadata in Libraries

Despite the benefits that ChatGTP offers in generating metadata in libraries, it also has its challenges. Ahmed et al., 2024 examined the role of metadata in promoting interoperability of AI-based prediction models. The study adopted systematic literature review. The study found that a major challenge in the application or use of metadata is that of lack of comprehensive metadata strategies to enhance both the interoperability and the compatibility of AI systems across different environments. (Noruzi, 2024) carried out a study on the use of artificial intelligence in knowledge organization and subject indexing. The result of his study showed that the degree of consistency in cataloguing and classification is low. A study by (Indraji et al., 2024) on the theme AI-powered digital tools for revolutionizing library operations: modern era posits that even though AI-powered metadata generation solutions save time as well as improve accuracy, freeing staff to focus on other important strategic objectives, underlying challenges associated with it are the issue of data security and privacy, biases in AI the systems and the caliber and variety of training data. (Oyighan et al., 2024) conducted a study on the role of AI in transforming metadata management: insights on challenges, opportunities, and emerging trends. The study adopted systematic literature review method. Their objectives were to explore the challenges libraries face with AI-driven metadata,

identify opportunities for improvement and analyze emerging trends. The findings from their study shows that ethical concerns, data quality and technical limitations were the major challenges associated with AI-powered metadata. The study concluded that human oversight, training of staff and updating policies related to AI-powered metadata is important in libraries and information centres.

Methodology

The study employed qualitative methodology with a case study technique to prompt ChatGPT to generate metadata for books. This was done in order to understand how ChatGPT would generate accurate and consistent metadata for information resources. ChatGPT version 4o released on May, 13th 2024 was queried to generate metadata in a plain text format and the Machine Readable Catalogue (MARC) formats of three books with the titles “Things fall Apart” by Chinua Achebe, "Library and Information Science Text for Africa" by L. O. Aina, and "Cryptocurrency, Blockchain, Ethereum, Bitcoin - The Complete Guide to Understanding Fintech" by George Icahn respectively. The responses were recorded. In order to test its consistency, the process was repeated after twenty four hours.

Results

Query 1a:

Please Provide metadata for the title:" Things Things Fall Apart" by Chinua Achebe

ChatGPT’s response:

Here is the metadata for the title *"Things Fall Apart"* by Chinua Achebe:

- Title: Things Fall Apart
- Author: Chinua Achebe
- Published: 1958
- Publisher: Heinemann

- ISBN: 9780435905255 (varies by edition)
- Format: Print (also available in eBook and audiobook formats)
- Pages: 209 pages (varies by edition)
- Genre: Fiction, African Literature, Historical Fiction
- Language: English
- Setting: Nigeria, late 19th century
- Subjects: African traditional society, colonialism, Igbo culture, clash of cultures, personal tragedy
- Keywords: Colonialism, Igbo society, tradition vs modernity, identity, cultural conflict, pre-colonial Africa

Things Fall Apart is one of the most famous African novels, portraying the impact of European colonization on traditional African societies, particularly the Igbo people in Nigeria. It remains a foundational text in post-colonial literature and African studies.

Query 1b:

Provide me with the MARC format:

ChatGPT's response:

=LDR 00000nam 2200000 a 4500

=001 9780435905255

=005 20240914124500.0

=008 580101s1958####enk##########000#1#eng##

=020 ##$a9780435905255

=040 ##$aTWI$beng$cTWI

=100 1#$aAchebe, Chinua,$eauthor.

=245 10$aThings fall apart /$cby Chinua Achebe.

=250 ##$a1st edition.

=260 ##$aLondon :$bHeinemann,$c1958.

=300 ##$a209 p. ;$c20 cm.

=336 ##$atext$btxt$2rdacontent

=337 ##$aunmediated$bn$2rdamedia

=338 ##$avolume$bnc$2rdacarrier

=490 0#$aAfrican Writers Series ;$v1

=500 ##$aFirst published in 1958.

=520 ##$aThis novel portrays the life of Okonkwo, an Igbo man, and the impact of British colonialism on his society in Nigeria during the late 19th century.

=650 #0$aIgbo (African people)$vFiction.

=650 #0$aColonialism$xAfrica$vFiction.

=650 #0$aNigeria$xSocial life and customs$vFiction.

=650 #0$aCultural conflict$vFiction.

=650 #0$aAfrica$xHistory$vFiction.

=700 1#$aAchebe, Chinua.

=830 #0$aAfrican Writers Series ;$v1

=999 ##$aHeinemann.

This is a standard MARC 21 record with field codes, offering bibliographic information for *Things Fall Apart*. It includes important fields like title (245), author (100), publication information (260), subject headings (650), and more.

Query 2a:

Please Provide metadata for the title "Library and Information Science Text for Africa" by L. O. Aina

ChatGPT's response:

Here is the metadata for the book *"Library and Information Science Text for Africa"* by L. O. Aina:

- Title: Library and Information Science Text for Africa
- Author: L. O. Aina
- Published: 2004
- Publisher: Third World Information Services Ltd.
- ISBN: 9789783561200
- Format: Print
- Pages: 441 pages
- Subject: Library and Information Science, Information Management.
- Keywords: Library science, information management, African libraries, library services, information technology in libraries.

This book serves as a comprehensive guide for students, educators, and professionals in library and information science, particularly focused on the African context.

Query 2b:

Provide me with the MARC format:

ChatGPT's response:

=LDR 00000nam 2200000 a 4500

=001 9789783561200

=005 20240914123456.0

=008 040621s2004####nr#a#########000#0#eng##

=020 ##$a9789783561200

=040 ##$aTWI$beng$cTWI

=100 1#$aAina, L. O.$q(Lawal Oladipupo),$eauthor.

=245 10$aLibrary and information science text for Africa /$cby L. O. Aina.

=260 ##$aIbadan, Nigeria :$bThird World Information Services Ltd.,$c2004.

=300 ##$a441 p. :$bill. ;$c25 cm.

=650 #0$aLibrary science$xAfrica.

=650 #0$aInformation science$xAfrica.

=650 #0$aInformation technology$xLibraries$xAfrica.

=650 #0$aLibrary management$xAfrica.

=700 1#$aAina, L. O.$q(Lawal Oladipupo)

=999 ##$aTWI$bThird World Information Services Ltd.

Query 3a:

Provide me with metadata for the book titled: "Cryptocurrency, Blockchain, Ethereum, Bitcoin - The Complete Guide to Understanding Fintech" by George Icahn:

ChatGPT's response 1:

Here is the metadata for the book *"Cryptocurrency, Blockchain, Ethereum, Bitcoin - The Complete Guide to Understanding Fintech"* by George Icahn:

- Title: Cryptocurrency, Blockchain, Ethereum, Bitcoin - The Complete Guide to Understanding Fintech
- Author: George Icahn
- Published: 2017
- Publisher: Createspace Independent Publishing Platform
- ISBN: 9781977931733
- Format: Print and eBook
- Pages: 92 pages

- Genre: Non-fiction, Technology, Finance
- Language: English
- Subjects: Cryptocurrency, Blockchain technology, Bitcoin, Ethereum, Fintech, Digital finance
- Keywords: Cryptocurrency, Blockchain, Bitcoin, Ethereum, financial technology, digital currency, decentralized finance

This book provides an introduction to financial technology (Fintech), specifically focusing on cryptocurrency and blockchain technology, with an emphasis on Bitcoin, Ethereum, and related digital currencies.

Query 3b

Provide me with metadata for the print version of the book titled: "Cryptocurrency, Blockchain, Ethereum, Bitcoin - The Complete Guide to Understanding Fintech" by George Icahn:

ChatGPT's response 3b:

Here is the metadata for the print format of *"Cryptocurrency, Blockchain, Ethereum, Bitcoin - The Complete Guide to Understanding Fintech"* by George Icahn:

- Title: Cryptocurrency, Blockchain, Ethereum, Bitcoin - The Complete Guide to Understanding Fintech
- Author: George Icahn
- Published: 2017
- Publisher: Createspace Independent Publishing Platform
- ISBN: 9781977931733
- Format: Print (Paperback)
- Pages: 92 pages
- Dimensions: 6 x 0.2 x 9 inches

- Language: English
- Genre: Non-fiction, Technology, Finance
- Subjects: Cryptocurrency, Blockchain technology, Bitcoin, Ethereum, Fintech, Digital finance
- Keywords: Cryptocurrency, Blockchain, Bitcoin, Ethereum, financial technology, digital currency, decentralized finance
- Edition: 1st Edition

This metadata is specific to the print (paperback) edition of the book. It provides essential details such as the book's physical dimensions and edition information

Query 3b:

Provide me with the MARC format:

=LDR 00000nam 2200000 a 4500

=001 9781977931733

=005 20240914130000.0

=008 171001s2017####xxu###########000#0#eng##

=020 ##$a9781977931733$q(paperback)

=040 ##$aTWI$beng$cTWI

=100 1#$aIcahn, George,$eauthor.

=245 10$aCryptocurrency, blockchain, Ethereum, Bitcoin :$bthe complete guide to understanding fintech /$cby George Icahn.

=250 ##$a1st edition.

=260 ##$a[United States] :$bCreatespace Independent Publishing Platform,$c2017.

=300 ##$a92 p. ;$c23 cm.

=336 ##$atext$btxt$2rdacontent

=337 ##$aunmediated$bn$2rdamedia

=338 ##$avolume$bnc$2rdacarrier

=500 ##$aSubtitle on cover: The Complete Guide to Understanding Fintech.

=520 ##$aThis book provides an introduction to the concepts of cryptocurrency, blockchain technology, and financial technology, focusing on Bitcoin, Ethereum, and other decentralized digital currencies.

=650 #0$aCryptocurrency.

=650 #0$aBlockchain (Databases).

=650 #0$aBitcoin.

=650 #0$aEthereum (Cryptocurrency).

=650 #0$aFinancial technology.

=700 1#$aIcahn, George.

=999 ##$aCreatespace Independent Publishing Platform.

Discussion of Findings

ChatGPT is a smart machine. When prompted to provide metadata for the books in queries one, two and three as seen above, it provided the metadata correctly without any misinformation or errors of spelling or any kind. This shows that ChatGPT provides metadata with accuracy, implying that it understands cataloguing rules. It even included a brief summary of the content of the book. Two of the three books are written by Nigerians in Nigeria. Even though, many Nigerian books have less visibility online, ChatGPT was able to learn about these and presented their metadata. This may mean that ChatGPT can learn about books and other information resources that are popular and this quality aids library and information professionals in carrying out tasks of creating metadata for library and information resources. Another interesting thing is that the machine is able to provide the Machine Readable Catalogue (MARC) format of the books which

means libraries using Library Management Systems can convert or directly upload the MARC records without necessarily creating them by themselves or copying from a library elsewhere. ChatGPT was queried the second time and it returned exactly the same result. This means that ChatGPT is consistent with the result it provides when it comes to metadata and MARC records. This finding agrees with (Wu et al., 2021) who posited that hold that unlike humans, who may introduce inconsistencies or errors, AI models can generate metadata that adheres to a consistent and free from typographical or semantic errors.

It is however, worthy of note that where a particular book has e-version, ChatGPT gives preference in providing metadata for the e-version ignoring that of the print version; this may be because the e-version is online compared to a print version that is offline. Again, ChatGPT do not provide call number as part of the metadata it presents and this implies that for call numbers, ChatGPT has to be queried separately. Also, ChatGPT provides limited metadata for some books compared with others. For example, in providing metadata for the book "Cryptocurrency, Blockchain, Ethereum, Bitcoin - The Complete Guide to Understanding Fintech" the machine included "Genre" and "Language" of the book and excluded them when it provided the metadata for the title: "Library and Information Science Text for Africa". This implies that ChatGPT may not have all the metadata elements for all the books or that it gives more preference to books that have e-versions online.

The challenges associated with use of ChatGPT in obtaining metadata for books are; network and inherent biases. ChatGPT return error message when prompted in an epileptic network environment; takes longer than the relatively usual time taken in displaying search result which affects the efficiency of ChatGPT. ChatGPT tends to be bias due to heavy online presence of a book. ChatGPT returns less comprehensive metadata for books with less online visibility thereby limiting level of detail when providing metadata for some books.This agrees with (Indraji

et al., 2024) that one of the challenges of AI-powered metadata is bias in the AI system associated with training data.

Conclusion

This study investigated the accuracy of AI-generated metadata, ascertained the consistency of AI in generating metadata and identified challenges associated with using AI-metadata. Based on the findings it is upheld that ChatGPT presents metadata with accuracy and consistency and lags or returns error message in an epileptic network environment. Therefore, there is great opportunity for librarians and other information professionals to leverage ChatGPT in creating metadata for information resources.

Recommendations

Based on the findings, the following recommendations are hereby made:

i. ChatGPT should be used in a strong and high bandwidth environment in order to ensure efficiency.
ii. A separate query should be made to ChatGPT when particularly in need for call number for a book.
iii. Books should be published in all formats including e-versions online so that ChatGPT can learn about them.

References

Ahmed, A. A., Abdullahi, A. U., Gital, A. Y., & Dutse, A. Y. (2024). The Role of Metadata in Promoting Explainability and Interoperability of AI-based Prediction Models. *Journal of Exceptional Multidisciplinary Research (JEMR)*, *1*(1), 33–45.

Cocciolo, A. (2025). Oral History Metadata and AI: A Study from an LGBTQ+ Archival Context. *Pratt Institute*.

Indraji, C., Dominic, J., & Kothandaraman, R. (2024). AI-Powered Digital Tools for Revolutionizing Library Operations: Modern Era. *South Indian Journal of Library and Information Science*, *2*(2), 1–10.

Kolawole, T. (2021). The impact of AI-generated metadata on information retrieval in libraries. Library Review, 70(6), 921-936.

López Espejel, G., García-Macias, J. A., & González-Caro, C. (2023). Evaluating the effectiveness of AI-generated metadata in library collections. *Journal of Information Science,* 49(3), 537-551.

Martins, S. (2024). Artificial Intelligence-Assisted Classification of Library Resources: The Case of Claude AI. *Library Philosophy and Practice (e-Journal)*, *8159*, 1–22. https://digitalcommons.unl.edu/libphilprac/8159

Musoke, M. (2022). AI-generated metadata: A study of its adoption and implementation in libraries. Library Management, 43(4), 247-263.

Noruzi, A. (2024). The Use of Artificial Intelligence in Knowledge Organization and Subject Indexing. *Infomology*, *3*(1), 1–8.

Olutoki, O., & Osoba, O. (2017). Metadata management in digital libraries: A review of the literature. *Journal of Librarianship and Information Science, 49(2), 155-171.*

Omame, I. M., & Alex-Nmecha, J. C. (2020). Artificial Intelligence in Libraries: In N. E. Osuigwe (Ed.), *Advances in Library and Information Science* (pp. 120–144). IGI Global. https://doi.org/10.4018/978-1-7998-1116-9.ch008

Oyighan, D., Ukubeyinje, E. S., David-West, B. T., & Oladokun, B. D. (2024). The Role of AI in Transforming Metadata Management: Insights on Challenges, Opportunities, and Emerging Trends. *Asian Journal of Information Science and Technology*, *14*(2), 20–26.

Park, J., & Lu, C. (2020). AI-generated metadata for digital resources: A survey of current practices. Journal of Digital Libraries, 11(2), 123-145.

Sonawane, S., & Chavan, U. (2020). AI-generated metadata: A systematic review of the literature. Journal of Information Technology, 30(1), 34-50.

Stepanov, A. (2019). The role of AI-generated metadata in improving discoverability in libraries. Library Hi Tech, 37(2), 257-275.

Thelwall, M. (2024). Can ChatGPT evaluate research quality? *Journal of Data and Information Science*, *9*(2), 1–21.

Wada, I., Digma, F. Z., & Joel, A. P. (2023). Artificial intelligence (AI) and machine learning for metadata generation and library service delivery: A systematic review. *Journal of Information Practice and Management, 3*(1). Retrieved from https://www.researchgate.net/publication/378373383

Williams, P. (2021). AI-generated metadata: Opportunities and challenges for libraries. Journal of Library Administration, 61(2), 155-172.

Yang, X., Wang, J., Zhang, J., & Li, W. (2019). AI-generated metadata for cultural heritage collections: A case study. *Journal of Cultural Heritage,* 38(2), 251-265.

Yusuf, A., et al. (2021). Evaluating the quality of AI-generated metadata in libraries. *Journal of Information Science, 47(4), 567-583.*

Author Note

Sugabsen Martins, Abdurrahman Ghaji Library, Adamawa State University, Mubi.

Matins1085@adsu.edu.ng, 07038015309

Ukawuba Vivian Chinazo, National Root Crops Research Institute, Umahia, Abia State.

vindunita@yahoo.com, 08037452017